

LAB ASSIGNMENT 1

QUESTION 1

Q1. Create a LabFile directory and perform create/copy/move/rename/delete operations using touch, cp, mv, rm, unlink, ls, stat.

- touch - Creates empty files if they don't already exist, or updates the modification timestamp of existing files.

Syntax: - touch file...

```
user$ echo "Hello World!"&clear
Hello World!&clear
user$ mkdir LabFiles
user$ cd LabFiles
user$ touch file1.txt file2.txt file3.txt
user$ ls
file1.txt
file2.txt
file3.txt
```

- cp - Copies a file to a new location or name, keeping the original untouched.

Syntax: - cp source dest

```
user$ cp file1.txt copy_file1.txt
user$ ls
copy_file1.txt
file1.txt
file2.txt
file3.txt
```

- mv - Moves or renames a file or directory.

Syntax: - mv source dest

```
user$ mv file2.txt moved_file2.txt
user$ ls
  copy_file1.txt
  file1.txt
  file3.txt
  moved_file2.txt
user$ mv file3.txt renamed_file3.txt
user$ ls
  copy_file1.txt
  file1.txt
  moved_file2.txt
  renamed_file3.txt
user$ rm renamed_file3.txt
user$ ls
  copy_file1.txt
  file1.txt
  moved_file2.txt
```

- rm - Removes files; -r removes directories recursively , -f suppresses confirmation prompts and missing-file errors.

Syntax: - rm [r] [f] file...

```
user$ rm renamed_file3.txt
user$ ls
  copy_file1.txt
  file1.txt
  moved_file2.txt
```

- unlink – Removes a single file or link — similar to a very simple rm with no options.
- Syntax: - unlink file

```
user$ touch deletefile.txt
user$ ls
copy_file1.txt
deletefile.txt
file1.txt
moved_file2.txt
user$ unlink deletefile.txt
user$ ls
copy_file1.txt
file1.txt
moved_file2.txt
```

- `ls` – Lists directory contents; `-l` gives a detailed long listing (permissions, owner, size, date), `-R` lists recursively.

Syntax: `ls [-l] [-R] [path]`

```
user$ ls
copy_file1.txt
file1.txt
moved_file2.txt
user$ ls -l
total 0
-rw-r--r-- 1 1066 1066 0 Sep 28 12:41 copy_file1.txt
-rw-r--r-- 1 1066 1066 0 Sep 28 12:41 file1.txt
-rw-r--r-- 1 1066 1066 0 Sep 28 12:41 moved_file2.txt
```

- `stat` – Displays detailed metadata about a file: size, permissions, timestamps and inode number.

Syntax: `stat file`

```
user$ stat file1.txt
  File: file1.txt
  Size: 0          Blocks: 0          IO Block: 4096   regular empty file
Device: 0,51    Inode: 1860403     Links: 1
Access: (0644/-rw-r--r--)  Uid: ( 1066/ UNKNOWN)   Gid: ( 1066/ UNKNOWN)
Access: 2026-09-28 12:42:07.922308870 +0000
Modify: 2026-09-28 12:42:07.898308869 +0000
Change: 2026-09-28 12:42:07.898308869 +0000
 Birth: 2026-09-28 12:42:07.898308869 +0000
```

Final Answer:

```
user$ mkdir LabFiles
user$ cd LabFiles
user$ touch file1.txt file2.txt file3.txt
user$ ls
  file1.txt
  file2.txt
  file3.txt
user$ cp file1.txt copy_file1.txt
user$ ls
  copy_file1.txt
  file1.txt
  file2.txt
  file3.txt
user$ mv file2.txt moved_file2.txt
user$ ls
  copy_file1.txt
  file1.txt
  file3.txt
  moved_file2.txt
```

```
user$ mv file3.txt renamed_file3.txt
user$ ls
  copy_file1.txt
  file1.txt
  moved_file2.txt
  renamed_file3.txt
user$ rm renamed_file3.txt
user$ ls
  copy_file1.txt
  file1.txt
  moved_file2.txt
user$ touch deletefile.txt
user$ ls
  copy_file1.txt
  deletefile.txt
  file1.txt
  moved_file2.txt
user$ unlink deletefile.txt
```

```
user$ ls
  copy_file1.txt
  file1.txt
  moved_file2.txt
user$ ls -l
total 0
-rw-r--r-- 1 1066 1066 0 Sep 28 12:41 copy_file1.txt
-rw-r--r-- 1 1066 1066 0 Sep 28 12:41 file1.txt
-rw-r--r-- 1 1066 1066 0 Sep 28 12:41 moved_file2.txt
user$ stat file1.txt
  File: file1.txt
  Size: 0          Blocks: 0          IO Block: 4096   regular empty file
Device: 0,51    Inode: 1860403    Links: 1
Access: (0644/-rw-r--r--)  Uid: ( 1066/ UNKNOWN)   Gid: ( 1066/ UNKNOWN)
Access: 2026-09-28 12:42:07.922308870 +0000
Modify: 2026-09-28 12:42:07.898308869 +0000
Change: 2026-09-28 12:42:07.898308869 +0000
 Birth: 2026-09-28 12:42:07.898308869 +0000
user$
```

LAB ASSESSMENT 1

QUESTION 2

Q2. Demonstrate pipes & redirection (>, >>, <, 2>, 2>>, 2>&1, |) with cat, grep, sort, uniq, wc, tee to generate filtered reports.

•cat – Concatenates and prints the entire contents of one or more files to the screen.

Syntax: - cat file...

```
user$ mkdir Pipelab
user$ cd Pipelab
user$ cat > student.txt << EOF
user$ adi mca
user$ chiki mca
user$ rinki mca
user$ tishu mca
user$ aman cse
user$ mihra cse
user$ joya cse
user$ eof
user$ EOF
user$ cat student.txt
adi mca
chiki mca
rinki mca
tishu mca
aman cse
mihra cse
joya cse
```

```

eof
user$ echo "student department report" > report.txt
user$ cat report.txt
student department report
user$ cat student.txt >> report.txt
user$ cat report.txt
student department report
adi mca
chiki mca
rinki mca
tishu mca
aman cse
mihra cse
joya cse
eof

```

- grep - Searches for lines matching a pattern; -i ignores case, -n prefixes match with their line number.

Syntax: - grep [-i] [-n] pattern file

```

user$ grep "cse" student.txt | tee cse_report.txt
aman cse
mihra cse
joya cse
user$ grep "mca" student.txt | tee -a mca_report.txt
adi mca
chiki mca
rinki mca
tishu mca

```

- sort - Sorts the lines of a file; -n sorts numerically, -r reverses the order, -t/-k choose a delimiter and field to sort by.

Syntax: - Sort [-t sep] [-k N] [-n] [-r] file

```

user$ sort < student.txt
adi mca
aman cse
chiki mca
eof
joya cse
mihra cse
rinki mca
tishu mca

user$ sort student.txt | uniq
adi mca
aman cse
chiki mca
eof
joya cse
mihra cse
rinki mca
tishu mca

```

- uniq - Removes consecutive duplicate lines from (typically already sorted) input.

Syntax: - uniq [file]

```

user$ sort student.txt | uniq
adi mca
aman cse
chiki mca
eof
joya cse
mihra cse
rinki mca
tishu mca

```

- wc - Counts lines (-l), words (-w) or characters (-c) in a file.

Syntax: - wc [-l] [-w] [-c] file

```

user$ wc -l student.txt
8 student.txt

user$ sort student.txt | uniq | wc -l
8

```

- tee - Reads from standard input and writes it to both standard output and a file at the same time.

Syntax: - cmd | tee file

```
user$ grep "cse" student.txt | tee cse_report.txt
aman cse
mihra cse
joya cse
user$ grep "mca" student.txt | tee -a mca_report.txt
adi mca
chiki mca
rinki mca
tishu mca
```

FINAL ANSWER

```
user$ mkdir Pipelab
user$ cd Pipelab
user$ cat > student.txt << EOF
user$ adi mca
user$ chiki mca
user$ rinki mca
user$ tishu mca
user$ aman cse
user$ mihra cse
user$ joya cse
user$ eof
user$ EOF
user$ cat student.txt
adi mca
chiki mca
rinki mca
tishu mca
aman cse
mihra cse
joya cse
```

```
eof
user$ echo "student department report" > report.txt
user$ cat report.txt
student department report
user$ cat student.txt >> report.txt
user$ cat report.txt
student department report
adi mca
chiki mca
rinki mca
tishu mca
aman cse
mihra cse
joya cse
eof
user$ grep "cse" student.txt | tee cse_report.txt
aman cse
mihra cse
joya cse
user$ grep "mca" student.txt | tee -a mca_report.txt
```

```
adi mca
chiki mca
rinki mca
tishu mca

user$ cat mca_report
cat: mca_report: No such file or directory

user$ cat cse_report.txt
aman cse
mihra cse
joya cse

user$ sort < student.txt
adi mca
aman cse
chiki mca
eof
joya cse
mihra cse
rinki mca
tishu mca

user$ sort student.txt | uniq
```

```
adi mca
aman cse
chiki mca
eof
joya cse
mihra cse
rinki mca
tishu mca

user$ wc -l student.txt
8 student.txt

user$ sort student.txt | uniq | wc -l
8

user$ |
```

LAB ASSESSMENT 1

QUESTION 3

Q3. Create College directory hierarchy with subdirectories (CSE, ESE, ME, Civil), nested semester/subject folders, and display it using tree, pwd, cd, mkdir, rmdir, ls.

- **mkdir** - Creates one or more new directories; -p also creates missing parent directories and does not error if the directory already exists.

Syntax: - mkdir [-p] dir....

```
user$ mkdir college
```

- **pwd** - Prints the full path of the current working directory.

Syntax: - pwd

```
user$ pwd
/tmp/user/home/college
```

- **cd** - Changes the current working directory; cd - returns to the previous directory.

Syntax: - cd [dir]

```
user$ cd college
```

- **rmdir** - Removes an empty directory; fails with an error if the directory still contains files..

Syntax: - rmdir dir

```
user$ mkdir temp
user$ rmdir temp
```

- **ls** - Lists directory contents; -l gives a detailed long listing (permissions, owner, size, date), -R lists recursively.

Syntax: - ls [-l] [-R] [path]

```
user$ ls
CSE
Civil
ECE
ME
user$ ls CSE
Sem1
Sem2
user$ |
```

- **tree** - Displays a directory and its contents as a visual, indented tree.

Syntax: - **tree [dir]**

```
user$ tree
.
├── CSE
│   ├── Sem1
│   │   └── {C,C++}
│   └── Sem2
│       └── {Java,DBMS}
├── Civil
│   └── Sem1
│       └── {Mathematics,Engineering}
├── ECE
│   ├── Sem1
│   │   └── {Digital,Electronics}
│   └── Sem2
│       └── {Communication,Microprocessor}
└── ME
    ├── Sem1
    │   └── {Thermodynamics,Mechanics}
    └── Sem2
        └── {CAD,Manufacturing}
```

FINAL ANSWER

```
user$ mkdir college
user$ cd college
user$ pwd
/tmp/user/home/college
user$ mkdir CSE ECE ME Civil
user$ mkdir -p CSE/Sem1/{C,C++}
user$ mkdir -p CSE/Sem2/{Java,DBMS}
user$ mkdir -p ECE/Sem1/{Digital,Electronics}
user$ mkdir -p ECE/Sem2/{Communication,Microprocessor}
user$ mkdir -p ME/Sem1/{Thermodynamics,Mechanics}
user$ mkdir -p ME/Sem2/{CAD,Manufacturing}
user$ mkdir -p Civil/Sem1/{Mathematics,Engineering}
user$ pwd
/tmp/user/home/college
```

```
user$ tree
```

```
.
├── CSE
│   ├── Sem1
│   │   └── {C,C++}
│   └── Sem2
│       └── {Java,DBMS}
├── Civil
│   └── Sem1
│       └── {Mathematics,Engineering}
├── ECE
│   ├── Sem1
│   │   └── {Digital,Electronics}
│   └── Sem2
│       └── {Communication,Microprocessor}
└── ME
    ├── Sem1
    │   └── {Thermodynamics,Mechanics}
    └── Sem2
        └── {CAD,Manufacturing}
```

19 directories, 0 files

```
user$ ls -R
```

..

CSE

Civil

ECE

ME

./CSE:

Sem1

Sem2

./CSE/Sem1:

{C,C++}

./CSE/Sem1/{C,C++}:

./CSE/Sem2:

{Java,DBMS}

./CSE/Sem2/{Java,DBMS}:

Sem1

./Civil/Sem1:

{Mathematics,Engineering}

./Civil/Sem1/{Mathematics,Engineering}:

./ECE:

Sem1

Sem2

./ECE/Sem1:

{Digital,Electronics}

./ECE/Sem1/{Digital,Electronics}:

./ECE/Sem2:

{Communication,Microprocessor}

./ECE/Sem2/{Communication,Microprocessor}:

```
./ME/Sem2:  
{CAD,Manufacturing}  
  
./ME/Sem2/{CAD,Manufacturing}:  
user$ cd CSE  
user$ pwd  
/tmp/user/home/college/CSE  
user$ cd ..  
user$ mkdir temp  
user$ rmdir temp  
user$ ls  
CSE  
Civil  
ECE  
ME  
user$ ls CSE  
Sem1  
Sem2  
user$
```

LAB ASSESSMENT 1

QUESTION 4

Q4. **Configure files permissions & ownership (symbolic/numeric, chmod, chgrp, umask) and verify with ls -l, including recursive updates.**

- chmod - Changes file permissions, using symbolic (e.g. u+x) or numeric (e.g. 755) notation.

Syntax: - chmod mode file

```
user$ chmod u+rw file3.txt
user$ ls -l
total 0
-rw-r--r-- 1 1087 1087 0 Sep 28 14:27 file1.txt
-rw-r--r-- 1 1087 1087 0 Sep 28 14:27 file2.txt
-rwxr--r-- 1 1087 1087 0 Sep 28 14:27 file3.txt
-rw-r--r-- 1 1087 1087 0 Sep 28 14:27 file4.txt
```

- chown - Changes the owning user (and optionally group) of a file; usually requires sudo.

Syntax: - chown user[:group] file

```
user$ chmod o+r file4.txt
user$ ls -l
total 0
-rw-rw-r-- 1 1087 1087 0 Sep 28 14:28 file1.txt
-rw-r--r-- 1 1087 1087 0 Sep 28 14:28 file2.txt
-rwxr--r-- 1 1087 1087 0 Sep 28 14:28 file3.txt
-rw-r--r-- 1 1087 1087 0 Sep 28 14:28 file4.txt
user$ chmod 754 file4.txt
user$ ls -l
total 0
-rw-rw-r-- 1 1087 1087 0 Sep 28 14:29 file1.txt
-rw-r--r-- 1 1087 1087 0 Sep 28 14:29 file2.txt
-rwxr--r-- 1 1087 1087 0 Sep 28 14:29 file3.txt
-rwxr-xr-- 1 1087 1087 0 Sep 28 14:29 file4.txt
```

- umask - Sets the default permission mask applied automatically to newly created files and directories.

Syntax: - umask [mask]

```
user$ umask  
0022
```

- ls - Lists directory contents; -l gives a detailed long listing (permissions, owner, size, date), -R lists recursively.

Syntax: - ls [-l] [-R] [path]

```
user$ ls -l  
total 0  
-rw-rw-r-- 1 1085 1085 0 Sep 28 14:31 file1.txt  
-rw-r--r-- 1 1085 1085 0 Sep 28 14:31 file2.txt  
-rwxr--r-- 1 1085 1085 0 Sep 28 14:31 file3.txt  
-rwxr-xr-- 1 1085 1085 0 Sep 28 14:31 file4.txt
```

FINAL ANSWER

```
user$ mkdir labfile
user$ cd labfile
user$ pwd
/tmp/user/home/labfile
user$ touch file1.txt file2.txt file3.txt file4.txt
user$ ls
file1.txt
file2.txt
file3.txt
file4.txt
user$ ls -l
total 0
-rw-r--r-- 1 1085 1085 0 Sep 28 14:26 file1.txt
-rw-r--r-- 1 1085 1085 0 Sep 28 14:26 file2.txt
-rw-r--r-- 1 1085 1085 0 Sep 28 14:26 file3.txt
-rw-r--r-- 1 1085 1085 0 Sep 28 14:26 file4.txt
user$ chmod u+rw file3.txt
user$ ls -l
total 0
```

```
-rw-r--r-- 1 1087 1087 0 Sep 28 14:27 file1.txt
-rw-r--r-- 1 1087 1087 0 Sep 28 14:27 file2.txt
-rwxr--r-- 1 1087 1087 0 Sep 28 14:27 file3.txt
-rw-r--r-- 1 1087 1087 0 Sep 28 14:27 file4.txt

user$ chmod g+rw file1.txt
user$ ls -l
total 0
-rw-rw-r-- 1 1087 1087 0 Sep 28 14:28 file1.txt
-rw-r--r-- 1 1087 1087 0 Sep 28 14:28 file2.txt
-rwxr--r-- 1 1087 1087 0 Sep 28 14:28 file3.txt
-rw-r--r-- 1 1087 1087 0 Sep 28 14:28 file4.txt

user$ chmod o+r file4.txt
user$ ls -l
total 0
-rw-rw-r-- 1 1087 1087 0 Sep 28 14:28 file1.txt
-rw-r--r-- 1 1087 1087 0 Sep 28 14:28 file2.txt
-rwxr--r-- 1 1087 1087 0 Sep 28 14:28 file3.txt
-rw-r--r-- 1 1087 1087 0 Sep 28 14:28 file4.txt

user$ chmod 754 file4.txt
user$ ls -l
```

```
user$ ls -l
total 0
-rw-rw-r-- 1 1087 1087 0 Sep 28 14:29 file1.txt
-rw-r--r-- 1 1087 1087 0 Sep 28 14:29 file2.txt
-rwxr--r-- 1 1087 1087 0 Sep 28 14:29 file3.txt
-rwxr-xr-- 1 1087 1087 0 Sep 28 14:29 file4.txt
user$ sudo chown user file1.txt
sh: sudo: not found
user$ umask
0022
user$ ls -l
total 0
-rw-rw-r-- 1 1085 1085 0 Sep 28 14:31 file1.txt
-rw-r--r-- 1 1085 1085 0 Sep 28 14:31 file2.txt
-rwxr--r-- 1 1085 1085 0 Sep 28 14:31 file3.txt
-rwxr-xr-- 1 1085 1085 0 Sep 28 14:31 file4.txt
user$ chmod o-w file1.txt
user$ |
```

```
user$ mkdir labfile
user$ cd labfile
user$ pwd
/tmp/user/home/labfile
user$ touch file1.txt file2.txt file3.txt file4.txt
user$ ls
file1.txt
file2.txt
file3.txt
file4.txt
user$ ls -l
total 0
-rw-r--r-- 1 1085 1085 0 Sep 28 14:26 file1.txt
-rw-r--r-- 1 1085 1085 0 Sep 28 14:26 file2.txt
-rw-r--r-- 1 1085 1085 0 Sep 28 14:26 file3.txt
-rw-r--r-- 1 1085 1085 0 Sep 28 14:26 file4.txt
user$ chmod u+rw file3.txt
user$ ls -l
total 0
```

LAB ASSESSMENT 1

QUESTION

5. Create and manage users/groups: useradd, adduser, passwd, usermod, userdel, groupadd, groupmod, groupdel, groups, id, whoami.

Step 1 – Create and verify the group
Commands used: `sudo groupadd devteam`, `getent group devteam`. The devteam group was already present from the initial execution. The groupadd command reported that it already existed, and getent confirmed the group.

```
tamanna parashar (tamanna) [Running] - Oracle VirtualBox
vboxuser@tamanna:~/college/LabFiles$ sudo groupadd devteam
[sudo] password for vboxuser:
vboxuser@tamanna:~/college/LabFiles$ sudo groupadd devteam
groupadd: group 'devteam' already exists
vboxuser@tamanna:~/college/LabFiles$ getent group devteam
devteam:x:1003:
```

Step 2 – Create the user
Commands used: `sudo useradd -m -g devteam testuser1`, `id testuser1`. We created testuser1 with a home directory and assigned devteam as the primary group. id confirmed the user's UID, GID and group membership.

```
vboxuser@tamanna:~/college/LabFiles$ sudo useradd -m -g devteam testuser1
vboxuser@tamanna:~/college/LabFiles$ id testuser1
uid=1002(testuser1) gid=1003(devteam) groups=1003(devteam)
```

Step 3 – Set the user password
Commands used: `sudo passwd testuser1`. We assigned a password to testuser1. The passwd command completed successfully and displayed the password-updated confirmation.

```
vboxuser@tamanna:~/college/LabFiles$ sudo passwd testuser1
New password:
BAD PASSWORD: The password fails the dictionary check - It is too simplistic/systematic
Retype new password:
passwd: password updated successfully
```

Step 4 – Modify the user's group membership
Commands used: `sudo usermod -aG sudo testuser1`, `id testuser1`. We added testuser1 to the sudo group as a secondary group. The id output confirmed membership in both devteam and sudo.

```
vboxuser@tamanna:~/college/LabFiles$ sudo usermod -aG sudo testuser1
vboxuser@tamanna:~/college/LabFiles$ id testuser1
uid=1002(testuser1) gid=1003(devteam) groups=1003(devteam),27(sudo)
```

Step 5 – Display group memberships
Commands used: `groups testuser1`. We used groups to display the groups associated with testuser1. The output confirmed membership in devteam and sudo.

```
vboxuser@tamanna:~/college/LabFiles$ groups testuser1
testuser1 : devteam sudo
```

Step 6 – Display the current user Commands used: whoami. We used whoami to display the username of the current terminal session. The output shows the currently logged-in user.

```
vboxuser@tamanna:~/college/LabFiles$ whoami  
vboxuser  
vboxuser@tamanna:~/college/LabFiles$ sudo userdel
```

Step 7 – Delete the test user and verify Commands used: sudo userdel -r testuser1, id testuser1. We removed testuser1 and its home directory using userdel -r. The final id check confirmed that the user no longer exists.

```
vboxuser@tamanna:~/college/LabFiles$ sudo userdel -r testuser1  
userdel: testuser1 mail spool (/var/mail/testuser1) not found  
vboxuser@tamanna:~/college/LabFiles$ id testuser1  
id: 'testuser1': no such user  
vboxuser@tamanna:~/college/LabFiles$
```

QUESTION 6

6. Generate a system/directory report using find, du, df, stat, wc, sort, head, tail, uname, hostname, history, free, ps, top with screenshots and summary.

Step 1 – Count files in the College hierarchy Commands used: find College -type f | wc -l >> system_report.txt. We counted all files inside the College hierarchy and appended the resulting count to system_report.txt.

```
vboxuser@tamanna:~$ find ~ -type f | head
/home/vboxuser/permission.txt
/home/vboxuser/.profile
/home/vboxuser/.bash_logout
/home/vboxuser/students.txt
/home/vboxuser/snap/prompting-client/common/.cache/desktop-runtime-date
/home/vboxuser/snap/prompting-client/common/.cache/immodules/immodules.cache
/home/vboxuser/snap/prompting-client/common/.cache/gio-modules/giomodule.cache
/home/vboxuser/snap/prompting-client/common/.cache/gdk-pixbuf-loaders.cache
/home/vboxuser/snap/prompting-client/222/.last_revision
/home/vboxuser/snap/prompting-client/222/.config/user-dirs.locale
```

Step 2 – Check College directory disk usage Commands used: du -sh College >> system_report.txt. We calculated the total human-readable disk usage of the College directory and appended the result to the report.

```
vboxuser@tamanna:~$ du -sh ~
25M   /home/vboxuser
```

Step 3 – Check filesystem capacity Commands used: df -h . >> system_report.txt. We displayed the filesystem capacity for the current mount in human-readable form and appended the result to the report.

```
vboxuser@tamanna:~$ df -h
Filesystem      Size  Used Avail Use% Mounted on
tmpfs           1.5G  2.1M  1.5G   1% /run
/dev/sda2       30G   6.5G   22G  24% /
tmpfs           3.7G   0  3.7G   0% /dev/shm
none            1.0M   0  1.0M   0% /run/credentials/systemd-journald.service
tmpfs           3.7G   0  3.7G   0% /tmp
none            1.0M   0  1.0M   0% /run/credentials/systemd-resolved.service
none            1.0M   0  1.0M   0% /run/credentials/getty@tty1.service
tmpfs           741M  40K  741M   1% /run/user/1000
vboxuser@tamanna:~$ du -sh ~
25M   /home/vboxuser
vboxuser@tamanna:~$ find ~ -type f | head
```

Step 4- Find text files Commands used: find College -name "*.txt" >> system_report.txt. We searched the College hierarchy for .txt files and appended the matching paths to the report.

```
vboxuser@tamanna:~$ sort system_report.txt
/dev/sda2       30G   6.5G   22G  24% /
Filesystem      Size  Used Avail Use% Mounted on
Linux tamanna 7.0.0-31-generic #31-Ubuntu SMP PREEMPT_DYNAMIC Sat Aug 1 04:26:38 UTC
none            1.0M   0  1.0M   0% /run/credentials/getty@tty1.service
none            1.0M   0  1.0M   0% /run/credentials/systemd-journald.service
none            1.0M   0  1.0M   0% /run/credentials/systemd-resolved.service
tamanna
tmpfs           1.5G  2.1M  1.5G   1% /run
tmpfs           3.7G   0  3.7G   0% /dev/shm
tmpfs           3.7G   0  3.7G   0% /tmp
tmpfs           741M  40K  741M   1% /run/user/1000
```

Step 5 – Display College metadata Commands used: stat College >> system_report.txt. We collected metadata for the top-level College directory, including permissions, timestamps, size and inode information

```

vboxuser@tamanna:~$ df -h
Filesystem      Size  Used Avail Use% Mounted on
tmpfs            1.5G  2.1M  1.5G   1% /run
/dev/sda2        30G   6.5G   22G  24% /
tmpfs            3.7G   0  3.7G   0% /dev/shm
none            1.0M   0  1.0M   0% /run/credentials/systemd-journald.service
tmpfs            3.7G   0  3.7G   0% /tmp
none            1.0M   0  1.0M   0% /run/credentials/systemd-resolved.service
tmpfs            1.0M   0  1.0M   0% /run/credentials/getty@tty1.service
vboxuser@tamanna:~$ du -sh ~
25M    /home/vboxuser
vboxuser@tamanna:~$ find ~ -type f | head

```

Step 6 – Collect system information Commands used: `uname -a >> system_report.txt`. We recorded kernel, operating-system and architecture information in the report.

```

tamanna
vboxuser@tamanna:~$ uname -a
Linux tamanna 7.0.0-31-generic #31-Ubuntu SMP PREEMPT_DYNAMIC Sat Aug 1 04:26:38 UTC 20
vboxuser@tamanna:~$ df -h
Filesystem      Size  Used Avail Use% Mounted on
tmpfs            1.5G  2.1M  1.5G   1% /run
/dev/sda2        30G   6.5G   22G  24% /
tmpfs            3.7G   0  3.7G   0% /dev/shm
none            1.0M   0  1.0M   0% /run/credentials/systemd-journald.service
tmpfs            3.7G   0  3.7G   0% /tmp
none            1.0M   0  1.0M   0% /run/credentials/systemd-resolved.service
tmpfs            1.0M   0  1.0M   0% /run/credentials/getty@tty1.service
vboxuser@tamanna:~$ du -sh ~
25M    /home/vboxuser

```

Step 7 – Display the hostname Commands used: `hostname >> system_report.txt`. We recorded the machine hostname in the consolidated report.

```

vboxuser@tamanna:~$ touch system_report.txt
vboxuser@tamanna:~$ hostname
tamanna
vboxuser@tamanna:~$ uname -a

```

Question 4

Configure file permissions and ownership.

Step 1 – Create permission.txt and verify the file We created an empty file named permission.txt using touch and used ls to verify that the file was created.

```
vboxuser@tamanna:~/PermissionDemo$ touch test.txt
vboxuser@tamanna:~/PermissionDemo$ ls -l
total 0
-rw-rw-r-- 1 vboxuser vboxuser 0 Sep 23 16:14 test.txt
```

Step 2 – View the initial file permissions We used ls -l permission.txt to view the initial permissions, owner, group and other file details.

```
vboxuser@tamanna:~/PermissionDemo$ ls -l
total 0
-rw-rw-r-- 1 vboxuser vboxuser 0 Sep 23 16:14 test.txt
```

Step 3 – Change permissions to 755 We used chmod 755 permission.txt to give the owner read, write and execute permissions, while giving the group and others read and execute permissions.

```
-rw-rw-r-- 1 vboxuser vboxuser 0 Sep 23 16:14 test.txt
vboxuser@tamanna:~/PermissionDemo$ chmod 755 test.txt
vboxuser@tamanna:~/PermissionDemo$ ls -l test.txt
-rwxr-xr-x 1 vboxuser vboxuser 0 Sep 23 16:14 test.txt
vboxuser@tamanna:~/PermissionDemo$ chmod u+x test.txt
vboxuser@tamanna:~/PermissionDemo$ chmod g-w test.txt
vboxuser@tamanna:~/PermissionDemo$ ls -l
total 0
-rwxr-xr-x 1 vboxuser vboxuser 0 Sep 23 16:14 test.txt
```

Step 4 – Change file ownership We used sudo chown vboxuser:vboxuser permission.txt to set the file owner and group to vboxuser.

```
-rwxr-xr-x 1 vboxuser vboxuser 0 Sep 23 16:14 test.txt
vboxuser@tamanna:~/PermissionDemo$ sudo chown $USER test.txt
sudo) password for vboxuser:
sorry, try again.
sudo) password for vboxuser:
vboxuser@tamanna:~/PermissionDemo$ groups
vboxuser adm cdrom sudo dip plugdev uucp lpnetconf
```